INCH-POUND
MIL-PRF-20/29F
24 March 2017
SUPERSEDING
MIL-PRF-20/29E
21 April 2011

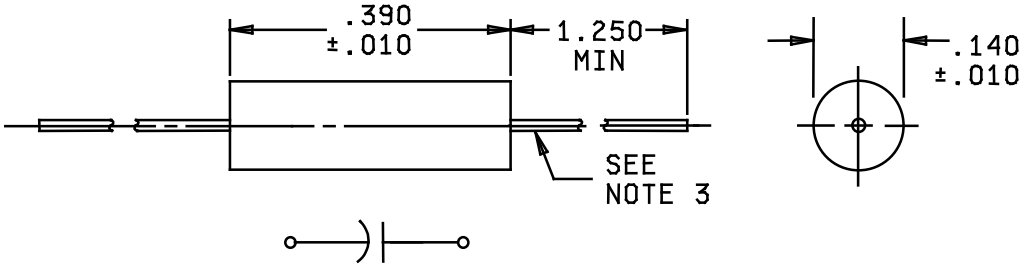
PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR77 AND CC77

CC77 is inactive for new design after 26 March 1984. Use CCR77

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-20.



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .015 inch (0.38 mm) to .027 inch (0.69 mm).
4. Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

Inches	mm
.010	0.25
.140	3.56
.390	9.91
1.250	31.75

FIGURE 1. Styles CCR77 and CC77 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR77 only) (percent per 1,000 hours): M (1.0), P (0.1), R (0.01) or S (0.001).

TABLE I. Capacitor characteristics.

Part or Identifying Number (PIN) 1/ 2/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC - 77CG151 - -	200	150	FGJ
CC - 77CG161 - -	200	160	FGJ
CC - 77CG181 - -	200	180	FGJ
CC - 77CG201 - -	200	200	FGJ
CC - 77CG221 - -	200	220	FGJ
CC - 77CG241 - -	200	240	FGJ
CC - 77CG271 - -	200	270	FGJ
CC - 77CG301 - -	200	300	FGJ
CC - 77CG331 - -	200	330	FGJ
CC - 77CG361 - -	200	360	FGJ
CC - 77CG391 - -	200	390	FGJ
CC - 77CG431 - -	200	430	FGJ
CC - 77CG471 - -	200	470	FGJ
CC - 77CG511 - -	200	510	FGJ
CC - 77CG561 - -	200	560	FGJ
CC - 77CG621 - -	200	620	FGJ
CC - 77CG681 - -	200	680	FGJ
CC - 77CG751 - -	100	750	FGJ
CC - 77CG821 - -	100	820	FGJ
CC - 77CG911 - -	100	910	FGJ
CC - 77CG102 - -	100	1000	FGJ
CC - 77CG112 - -	100	1100	FGJ
CC - 77CG122 - -	100	1200	FGJ
CC - 77CG132 - -	100	1300	FGJ
CC - 77CG152 - -	100	1500	FGJ
CC - 77CG162 - -	100	1600	FGJ
CC - 77CG182 - -	100	1800	FGJ
CC - 77CG202 - -	100	2000	FGJ
CC - 77CG222 - -	100	2200	FGJ
CC - 77CG242 - -	50	2400	FGJ
CC - 77CG272 - -	50	2700	FGJ
CC - 77CG302 - -	50	3000	FGJ

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

PIN 1/ 2/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC - 77CG332 - -	50	3300	FGJ
CC - 77CG362 - -	50	3600	FGJ
CC - 77CG392 - -	50	3900	FGJ
CC - 77CG432 - -	50	4300	FGJ
CC - 77CG472 - -	50	4700	FGJ
CC - 77CG512 - -	50	5100	FGJK
CC - 77CG562 - -	50	5600	FGJK

1/ Complete PIN will include the following:
1st dash - Symbol “R” (for style CCR77) or dash will be deleted (for style CC77).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR77 only) or dash will be deleted (for style CC77).

2/ Capacitors deleted since previous revisions shall be replaced by selecting a capacitor with the nearest parameter.

Thermal shock and voltage conditioning (CCR77 only): In accordance with MIL-PRF-20.

Capacitance: Within tolerance specified (see table I).

Marking: Method I of MIL-STD-1285, as shown in the following example:

ER

PIN	CCR77C G151FM
“J” brand and CAGE code	J12345
Date code and lot symbol (lot symbol may require two characters)	1120A

NON-ER

PIN	CC77C G151F
CAGE code	12345
Date code and lot symbol (lot symbol may require two characters)	1120A

Referenced documents. In addition to MIL-PRF-20, this specification sheet references the following document:

MIL-STD-1285

Changes from previous issue: The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2017-008)

Review activities:
Navy - AS, MC, OS, SH
Air Force – 99

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